

THE CLEBSCH CUBIC
RECOVERING, ORIENTING, AND REALIZING — III

Golden descent and operator realizations of the Clebsch cubic

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*From deep holes, the cubic takes shape, finds its bearings, and **stands fixed while its shadows move.***

Abstract

Hitchin’s degree-two incidence cover of harmonic cubics has rational square class $5J_0$. Thus its golden orientation is defined over $\mathbf{Q}(\sqrt{5})$, and the Clebsch chart satisfies $J_0 = 16\sigma_3^2$. After fixing the full marked bridge datum, we transport the chosen sign to the unique regular two-graph on six points and its conference operator C , with $C^2 = 5I$. Its balanced cross-block singular spectrum is cut-independent only in the trivial case and at order six, where it is $\{1/5, 4/5, 4/5\}$. In higher orders the exchange trace is universal, while the mean and variance of the second exchange moment are fixed by the classical 3-design of principal four-subsets with determinant -3 ; when marked, that design also reconstructs the signing up to switching and global negation. The resulting outer family of cubic forms has four equivalent descriptions: triangle holonomy, diagonal coefficients of $*\bigwedge^3 C$, commutator Pfaffians $\text{Pf}[D_x, C]$, and cross-golden determinants. Its six coordinates give the classical Joubert map to the Segre cubic; centered determinants give the Segre–Igusa polar map, and a distinguished hyperplane section is the diagonal Clebsch cubic surface.

The harmonic return uses the same Clebsch invariant. Degree-six zonal harmonics on the ten icosahedral face axes contain the Clebsch four-space as their Petersen eigenspace, and their normalized spherical cubic restricts as

$$\frac{1}{4\pi} \int_{S^2} F_y^3 = -\frac{784000}{1247103} \sigma_3(y).$$

The source-to-operator and operator-to-harmonic comparisons remain relative to an ordered golden realization, normalized chart lift, and compatible outer and Petersen labels. The incidence sheet alone does not recover these markings, and no rational or integral identification of the two ambient harmonic representations is asserted.

Keywords. Clebsch configuration; Hitchin cover; icosahedral invariant; arithmetic double cover; spherical harmonics; Gaunt invariant; symmetric conference matrix.

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1 Introduction

Let

$$H = \ker(\partial_x^2 + \partial_y^2 + \partial_z^2 : \mathbf{Q}[x, y, z]_3 \longrightarrow \mathbf{Q}[x, y, z]_1)$$

be the rational seven-space of harmonic cubics for $Q(x, y, z) = x^2 + y^2 + z^2$. We use the polarization pairing on cubics normalized by

$$\langle p, (a \cdot x)^3 \rangle = p(a);$$

its restriction to H is nondegenerate. Hitchin associated to a general $[f] \in \mathbf{P}(H)$ two icosahedral six-axis configurations: equivalently, two Mukai–Umemura isotropic three-planes $U \subset H$ with $f \perp U$ [1, 2]. The incidence variety of pairs $([f], U)$ is therefore generically a degree-two cover of $\mathbf{P}(H)$. We determine its rational square class and, after fixing the marked bridge datum below, compare its chosen sign with two further realizations. On the six-axis carrier the marked sign determines a conference operator whose Pfaffian, determinant, and middle-exterior shadows give the classical Joubert, Segre, Igusa, and diagonal Clebsch models. On the ten face axes it returns as the degree-six Gaunt cubic.

Let J_0 denote Hitchin’s irreducible invariant sextic, with the normalization in the displayed definition immediately preceding his main theorem and the matching final formula in his appendix [1, opening theorem and Appendix, final displayed formula]. Write

$$V = \{(y_1, \dots, y_5) : \sum_i y_i = 0\}, \quad \sigma_3(y) = \frac{1}{3} \sum_i y_i^3$$

for the Clebsch four-space and its invariant cubic. Here and below, $D(\sigma_3)$ is the principal open set $\{\sigma_3 \neq 0\}$. Because J_0 is a section of $\mathcal{O}_{\mathbf{P}(H)}(6)$, we use

$$K(\sqrt{cJ_0}), \quad K = \mathbf{Q}(\mathbf{P}(H)),$$

as shorthand: choose a nonzero rational section s of $\mathcal{O}_{\mathbf{P}(H)}(3)$, put $j_s = J_0/s^2 \in K^\times$, and take $K(\sqrt{cj_s})$. Replacing s changes j_s by a square, so the quadratic extension is unchanged.

Theorem 1.1 (Arithmetic incidence cover). *The function field of Hitchin’s incidence cover is*

$$\mathbf{Q}(\mathbf{P}(H))(\sqrt{5J_0}).$$

If $\mathcal{N} \rightarrow \mathbf{P}(H)$ is its finite Stein normalization, then, after normalizing the anti-invariant summand, its algebra is

$$\mathcal{O} \oplus \mathcal{O}(-3), \quad z^2 = 5J_0.$$

For the Clebsch chart $\iota_t : \mathbf{P}(V_{\mathbf{Q}(\sqrt{5})}) \rightarrow \mathbf{P}(H_{\mathbf{Q}(\sqrt{5})})$ defined in Section 2,

$$\iota_t^* J_0 = 16\sigma_3^2.$$

The chart is not defined over $\mathbf{Q}$; Galois conjugation carries it to the chart of the conjugate icosahedron. The fibre over the rational point $[xyz]$ is nevertheless a complete reduced fibre with residue algebra $\mathbf{Q}(\sqrt{5})$.

The underlying Galois exchange of the two conjugate icosahedra is classical [13, Section 1.3]. The theorem identifies its occurrence here with the square class of Hitchin’s incidence cover.

The factor 5 has a concrete descent meaning: the cubic $[xyz]$ and its unordered geometric fibre are rational, but choosing either icosahedral configuration—equivalently, either conjugate Clebsch chart through that point—requires $\mathbf{Q}(\sqrt{5})$.

The choice can be retained globally after normalizing the pullback to the Clebsch chart, but its comparison with the conference and harmonic signs is relative to a marking. A *marked bridge datum* $\mathbf{m}$ consists of an ordering of the six golden projective axes, the five plane triples labelled so that $q_1 = xyz$ and $\sum q_i = 0$, and the resulting normalized linear lift

$$\iota_t(y) = \sum_i y_i q_i.$$

The datum also identifies the ten face axes with the two-subsets of those five labels. Choose normalized representatives of the six projective axes. They give a conference matrix $C_{\mathbf{m}}$.

A different choice switches it by a diagonal sign matrix. The resulting switching-invariant triangle cubic is

$$Z_{\mathbf{m}}(x) = \sum_{i < j < k} (C_{\mathbf{m}})_{ij} (C_{\mathbf{m}})_{jk} (C_{\mathbf{m}})_{ki} x_i x_j x_k.$$

It is translation-invariant and hence defines

$$Z_{\mathbf{m}} \in \text{Sym}^3((\mathbf{Q}^6/\mathbf{Q}\mathbf{1})^*).$$

Write $[C_{\mathbf{m}}, Z_{\mathbf{m}}]_{\text{or}}$ for this marked source. Changing the auxiliary representatives by $a_i \mapsto d_i a_i$ sends $C_{\mathbf{m}}$ to $DC_{\mathbf{m}}D$ and fixes $Z_{\mathbf{m}}$. The opposite source is represented by $(-C_{\mathbf{m}}, -Z_{\mathbf{m}})$. The sign of the cubic generator is retained, so the brackets are not projective notation. Relabelling is equivariant only when all labelled objects are transported together; it is not quotiented silently.

Proposition 1.2 (Relative marked orientation bridge). *Fix a marked bridge datum $\mathbf{m}$. Let $B = \mathbf{P}(V_E)$, where $E = \mathbf{Q}(\sqrt{5})$, and pull Hitchin's incidence normalization back along $\iota_t : B \rightarrow \mathbf{P}(H_E)$. The normalization of this pullback is a disjoint union*

$$B_+ \amalg B_-.$$

The fixed isotropic plane selects B_+ , and incidence deck exchange interchanges the components. For an odd generator normalized on B_+ , their equations are

$$z = 4\sqrt{5}\sigma_3, \quad z = -4\sqrt{5}\sigma_3.$$

On $D(\sigma_3)$ they are the two distinct icosahedral configurations; on the Clebsch cubic the unnormalized branches meet, while the normalization remains disjoint.

Relative to $\mathbf{m}$, equip B_+ with the source $[C_{\mathbf{m}}, Z_{\mathbf{m}}]_{\text{or}}$ and equip B_- with its deck-opposite. At $[xyz]$, the golden exchanger identifies this opposite with the conjugate configuration and the representative $(-C_{\mathbf{m}}, -Z_{\mathbf{m}})$. Under the marked primitive pair-sum map

$$\beta : V \longrightarrow \mathbf{Q}^{\binom{5}{2}}, \quad \beta(y)_{ij} = y_i + y_j,$$

the same relative convention fixes the sign of the degree-six cubic in Theorem 6.1. The normalized component alone is not asserted to determine $\mathbf{m}$.

Here $Z_{\mathbf{m}}$ and σ_3 are different cubics on spaces of dimensions five and four. The proposition compares their orientations relative to $\mathbf{m}$; it does not identify their polynomial domains.

The operator stage is Theorem 5.3. The six outer translates of $Z_{\mathbf{m}}$ are simultaneously triangle cubics, middle-exterior diagonals, commutator Pfaffians, and cross-golden determinants. Their Joubert coordinates land on the Segre cubic, whose coordinate sections are diagonal Clebsch cubic surfaces; centered squaring is the classical Segre–Igusa polar map [13, Sections 2.1–2.2]. The marked datum defines the operator family. Before recognizing its four cubic shadows, Theorem 5.1 shows that order six is the unique nontrivial symmetric conference order whose balanced exchange spectrum is independent of the cut. At higher orders the classical determinant- (-3) design still fixes the mean and variance of the first nonconstant exchange moment. As a secondary reverse consequence, Theorem 5.2 shows that from order ten onward the marked design recovers the conference signing up to switching and global negation, and a quadratic family of selected determinant tests already suffices. This operator stage is the structural passage between the arithmetic source and the harmonic theorem below.

The arithmetic proof has three steps. Hitchin's degree and branch-sextic calculation reduces the generic extension to $K(\sqrt{cJ_0})$; the complete etale fibre at $[xyz]$, with residue

algebra $\mathbf{Q}(\sqrt{5})$, determines $c = 5$; and the explicit projective orthogonal transformation between the two configurations yields spinor class [2]. Independently, the harmonic proof identifies the pair-sum coefficients with the Petersen (-2) -eigenspace and determines the invariant cubic from one exact spherical-moment calculation. Between these arguments, Proposition 1.2 normalizes the chart pullback and, relative to the stated marking and lift, follows its deck involution through the golden conference matrix and Petersen pair-sum map.

Hitchin proves the generic degree, identifies the branch sextic, constructs the Clebsch chart, computes its restriction, and classifies the fibre over xyz [1, §§2–4, 7–10] [2, §§4–5]. The arithmetic theorem fixes the remaining rational constant in that square class and computes the spinor class of the explicit specialization. The Mukai–Umemura threefold supplies the original threefold [4]; Hitchin supplies the incidence construction and every formula used here. Dye’s finite theorem gives the complementary square-5 criterion for Clebsch hexagons [3]. The abstract equation $z^2 = 5J_{\mathbf{Z}}$ is étale on $D(10J_{\mathbf{Z}})$, but its comparison with the geometric incidence variety is proved only after inverting an unspecified integer.

The numerical harmonic restriction is logically independent of the incidence cover. The relative orientation theorem compares their signs only after choosing a sheet and the complete marked bridge datum. It places the classical degree-six bond-order observable—the rotationally invariant cubic formed from degree-six harmonic coefficients— [6, 7] on the same four-space and computes its exact restriction. The coefficient factors as

$$-\frac{784000}{1247103} = -\frac{400}{46189} \frac{1960}{27}.$$

The first factor is the universal $(6, 6, 6)$ Gaunt coupling $\left(\begin{smallmatrix} 6 & 6 & 6 \\ 0 & 0 & 0 \end{smallmatrix}\right)^2$; the second is the restriction scalar of the marked Petersen embedding. Thus the prime 11 in the denominator belongs to the universal angular momentum normalization, not to the golden descent. This paper relates the two constructions through their four-dimensional A_5 -module and invariant polynomial σ_3 , rather than by a rational map between their ambient harmonic spaces.

The decomposition of spherical harmonics under icosahedral symmetry is classical, including the degree-six symmetry types used here [9, 10]; modern applications likewise treat degree six as the lowest nonconstant even icosahedral channel [11]. Our calculation fixes a particular labelled ten-face-axis zonal map, identifies its Petersen four-space, and evaluates the cubic on that map. It does not claim priority for the ambient icosahedral decomposition or for the bond-order invariant.

2 The rational incidence cover

2.1 The Clebsch chart

Put $E = \mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5})$, and let

$$I_t = \{[0 : t : 1], [0 : t : -1], [1 : 0 : t], [-1 : 0 : t], \\ [t : -1 : 0], [-t : -1 : 0]\} \subset \mathbf{P}_E^2.$$

These are the six axes of the standard icosahedron. Define the linear subspace

$$V_t = \{p \in H_E : p(a) = 0 \text{ for every } [a] \in I_t\} \subset H_E.$$

This is the exact inclusion used below. Hitchin proves that V_t has dimension four and describes it as follows [1, §§2–4]. The rotation group of I_t permutes the five triples of

mutually orthogonal symmetry planes. Let $q_1, \dots, q_5$ be the products of the three linear equations in those triples, with $q_1 = xyz$ and relative scalars chosen so that $\sum_i q_i = 0$. Then

$$\iota_t : V_E = \{(y_i) \in E^5 : \sum_i y_i = 0\} \longrightarrow V_t, \quad (y_i) \longmapsto \sum_i y_i q_i$$

is an A_5 -equivariant isomorphism. Thus the parameter of $xyz = q_1$ is

$$y^\circ = \frac{1}{5}(4, -1, -1, -1, -1), \quad \sigma_3(y^\circ) = \frac{4}{25}.$$

With the polarization and sextic normalization fixed in the introduction, Hitchin's invariant calculation gives

$$\iota_t^* J_0 = 16\sigma_3^2, \quad J_0(xyz) = \left(\frac{16}{25}\right)^2 [1, \textit{Appendix}, \textit{finaldisplayedformula}].$$

The coefficient is fixed by the chosen lift, not by a further geometric parameter: at y° , the ratio of $J_0(xyz)$ to $\sigma_3(y^\circ)^2$ is exactly 16.

This chart is defined over E , not over $\mathbf{Q}$. The nontrivial automorphism $t \mapsto 1-t$ carries I_t and V_t to the distinct conjugate icosahedron and its vanishing space; the matrix R in Section 4 gives the corresponding transport. Hence ι_t is not a literal rational inclusion $V \hookrightarrow H$. The two displayed icosahedra have no common axis. Hitchin's two-icosahedron intersection theorem therefore gives

$$V_t \cap V_{1-t} = E \cdot xyz[1, \textit{Proposition 7}].$$

Consequently the reduced union $\mathbf{P}(V_t) \cup \mathbf{P}(V_{1-t})$ descends to $\mathbf{Q}$: over E it is a pair of conjugate projective three-spaces meeting only at the rational point $[xyz]$. Denote the descended scheme by Z . Its normalization map is $\nu : Z^\nu \rightarrow Z$, where $Z^\nu = \mathbf{P}(V_t)$, regarded as a $\mathbf{Q}$ -scheme through $\text{Spec } E \rightarrow \text{Spec } \mathbf{Q}$; after base change to E , this normalization is the disjoint union of the two conjugate components. The inverse image of the unique intersection point is

$$\nu^{-1}([xyz]) = \text{Spec } E.$$

The local algebra is an instance of a general pinching construction.

Quadratic pinching.

Let E/k be a separable quadratic field extension, put $A = E[[x_1, \dots, x_n]]$ with $n \geq 1$, and let $\mathfrak{m}$ be its maximal ideal. The fibre product

$$R = A \times_E k = k + \mathfrak{m}$$

has normalization A , conductor $\mathfrak{m}$, and defect $A/R \simeq E/k$ as a k -vector space. After completed base change to E , it is the union of two formal n -planes meeting at their origins.

Choose $\alpha \in E \setminus k$. Then $A = R + R\alpha$, so A is finite, hence integral, over R . The two rings have the same fraction field because $\alpha = (\alpha x_1)/x_1$. Since A is normal, it is the normalization. An element of the conductor first lies in R , so its residue is some $c \in k$. Multiplication by the constants in E forces $cE \subset k$, hence $c = 0$; conversely $\mathfrak{m}A \subset \mathfrak{m}$. Thus the conductor is $\mathfrak{m}$, and reduction modulo $\mathfrak{m}$ gives $A/R \simeq E/k$ as k -vector spaces. Finally $E \otimes_k E \simeq E \times E$; the fibre-product description gives the two conjugate branches after completed base change.

The two projective three-spaces above meet transversely, so the completed local ring of Z is this pinching ring with $k = \mathbf{Q}$, $E = \mathbf{Q}(\sqrt{5})$, and $n = 3$. Its normalization, conductor, and defect $E/\mathbf{Q}$ are therefore forced by residue-field pinching; they do not depend on a six-variable presentation. Thus 5 is the discriminant square class of the two branches, not

merely a value recovered from the sextic. This local model explains the descent geometry but does not replace the comparison with Hitchin's incidence scheme below. After base change to E , the pullback of the incidence double cover to $D(\sigma_3)$ is split. Galois conjugation exchanges both the two components and the two conjugate charts.

2.2 A rational incidence model

The function-field calculation requires a rational model, which can be given without descending either Clebsch chart. Let $W = \mathbf{Q}^3$ with the quadratic form Q from the introduction. For $a \in W$, infinitesimal rotation about a acts on H ; put

$$\omega_a(p, q) = \langle a \cdot p, q \rangle.$$

These three rational skew forms define a closed subscheme

$$X = \{U \in \mathrm{Gr}(3, H) : \omega_a|_U = 0 \text{ for every } a \in W\}.$$

Hitchin identifies $X_{\mathbf{C}}$ with the Mukai–Umemura threefold [2, §§4–5]; in particular X is geometrically integral and smooth. The incidence scheme used here is

$$\mathcal{I} = \{([f], U) \in \mathbf{P}(H) \times X : \langle f, U \rangle = 0\}.$$

Let $\pi : \mathcal{I} \rightarrow \mathbf{P}(H)$ be the first projection. It is a projective bundle over X , is therefore normal and integral, and all its displayed equations are defined over $\mathbf{Q}$. Mukai and Umemura introduced the underlying threefold; Hitchin supplies this Grassmannian model and the incidence calculation [4, 2].

There is also a canonical link from the chart model to the finite incidence cover. Let

$$\mathcal{N} = \mathrm{Spec}_{\mathbf{P}(H)} \pi_* \mathcal{O}_{\mathcal{I}}.$$

Stein factorization gives $\mathcal{I} \rightarrow \mathcal{N} \rightarrow \mathbf{P}(H)$. Since $\mathcal{I}$ is normal and the degree calculation below makes its generic field quadratic over $\mathbf{Q}(\mathbf{P}(H))$, $\mathcal{N}$ is the normalization of $\mathbf{P}(H)$ in that field. For the icosahedron I_t , let $U_t \in X(E)$ be Hitchin's isotropic three-plane. The polarization identity and Hitchin's description give [1, §§2–5]

$$V_t = U_t^\perp,$$

so $[p] \mapsto ([p], U_t)$ defines a section $\mathbf{P}(V_t) \rightarrow \mathcal{I}_E$. The two conjugate sections descend to a morphism

$$Z^\nu \longrightarrow \mathcal{I} \longrightarrow \mathcal{N}$$

over $\mathbf{P}(H)$. Over the finite etale neighborhood of $[xyz]$ constructed below, the local comparison proves that this morphism carries $\nu^{-1}([xyz]) = \mathrm{Spec} E$ isomorphically onto the golden incidence fibre.

2.3 Proof of Theorem 1.1

Hitchin realizes the Mukai–Umemura threefold as the zero locus of the three skew forms on $\mathrm{Gr}(3, H)$. The top Chern number of the dual universal bundle is 2, so a general f is orthogonal to two isotropic three-planes [2, §§4–5]. The incidence variety is a projective bundle over the integral Mukai–Umemura threefold, hence is integral; its generic function field is therefore a quadratic extension.

Square class from the branch divisor. Write $K = \mathbf{Q}(\mathbf{P}(H))$. In characteristic zero every quadratic extension of K has the form $K(\sqrt{d})$. The prime divisors at which d has odd

valuation are exactly the branch divisors. Hitchin identifies the branch hypersurface with the irreducible invariant sextic $J_0 = 0$ [1, §§9–10].

Choose a nonzero rational section s of $\mathcal{O}_{\mathbf{P}(H)}(3)$ and set

$$j_s = \frac{J_0}{s^2} \in K^\times.$$

Its odd valuations occur exactly along $J_0 = 0$, because the divisor of s^2 is even. Consequently

$$\operatorname{div}(d/j_s) = 2D$$

for a divisor D on $\mathbf{P}(H)$. The class of D is two-torsion. Since $\operatorname{Pic}(\mathbf{P}(H)) = \mathbf{Z}$, D is principal. Thus

$$d = c j_s g^2$$

for some $g \in K^\times$ and a unique $c \in \mathbf{Q}^\times/\mathbf{Q}^{\times 2}$. The incidence extension is therefore $K(\sqrt{cJ_0})$.

The local comparison at xyz . Hitchin's classification gives exactly the two distinct configurations I_t and I_{1-t} over $[xyz]$ [1, §§3, 7–9]. Thus π is quasi-finite over this point. Because π is proper, there is a Zariski neighborhood B of $[xyz]$ on which π is finite. Let $\tilde{B}$ be the normalization of B in the quadratic generic field $K(\sqrt{cJ_0})$. The normal finite scheme $\pi^{-1}(B)$ is birational to $\tilde{B}$, hence is isomorphic to it. Hitchin's branch calculation identifies the branch locus with $J_0 = 0$. Since $J_0(xyz) \neq 0$, shrinking B away from this divisor makes the finite normalization étale. The two distinct geometric points therefore form the complete reduced scheme-theoretic fibre, whose residue algebra is

$$E = \mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5}).$$

Choose a local generator e of $\mathcal{O}_{\mathbf{P}(H)}(3)$ at $[xyz]$, and write $j_e = J_0/e^2$. The normalization is locally $w^2 = c j_e$. Changing e multiplies j_e by a unit square. Moreover, $j_e(xyz)$ differs from the displayed value $J_0(xyz) = (16/25)^2$ by a rational square. Specialization therefore gives the residue algebra $\mathbf{Q}(\sqrt{c})$. Comparison with the preceding fibre proves $c = 5$ in $\mathbf{Q}^\times/\mathbf{Q}^{\times 2}$. This also explains why evaluating the generic square class at the point is legitimate: the point lies in the finite étale locus and the change from j_s to j_e is a unit square there.

The Stein algebra. The quadratic field involution extends to the normalization $\mathcal{N}$. If $f : \mathcal{N} \rightarrow \mathbf{P}(H)$, then, since 2 is invertible,

$$f_*\mathcal{O}_{\mathcal{N}} = \mathcal{O} \oplus \mathcal{M}$$

is the decomposition into invariant and anti-invariant summands. The invariant summand is $\mathcal{O}$: its relative spectrum is finite and birational over the normal space $\mathbf{P}(H)$. Normality makes $f_*\mathcal{O}_{\mathcal{N}}$ an S_2 module over the regular base, and the direct summand $\mathcal{M}$ is therefore a rank-one reflexive sheaf. Every such sheaf on projective space is a line bundle, say $\mathcal{M} \simeq \mathcal{O}(-d)$.

Multiplication on the anti-invariant summand is a section of $\mathcal{M}^{-2} \simeq \mathcal{O}(2d)$. Its zero divisor is the branch divisor, which is the sextic $J_0 = 0$; hence $d = 3$, and multiplication is cJ_0 for some $c \in \mathbf{Q}^\times$. The function-field calculation and the golden fibre give $c = 5$ modulo a rational square. Rescaling the chosen isomorphism $\mathcal{M} \simeq \mathcal{O}(-3)$ absorbs that square and gives the exact algebra

$$f_*\mathcal{O}_{\mathcal{N}} \simeq \mathcal{O} \oplus \mathcal{O}(-3), \quad z^2 = 5J_0.$$

After Hitchin's scale for J_0 and an isomorphism $\mathcal{M} \simeq \mathcal{O}(-3)$ giving multiplication $5J_0$ are fixed, the remaining choices differ by $z \mapsto -z$, the deck involution. Thus the equation used on the Clebsch chart below is an equation of finite algebras, not only of their function fields.

Integral boundary. Choose a full lattice $\Lambda \subset H$ and an integer $a \neq 0$ such that $J_{\mathbf{Z}} = a^2 J_0$ has integral coefficients. The algebra

$$\mathcal{O}_{\mathbf{P}(\Lambda)} \oplus \mathcal{O}_{\mathbf{P}(\Lambda)}(-3), \quad z^2 = 5J_{\mathbf{Z}},$$

is finite locally free of degree two over $\mathbf{Z}$ and has the function field of Theorem 1.1. On $D(10J_{\mathbf{Z}})$ it is finite etale. The primes 2 and 5 are forced bad primes for an ordinary two-sheet interpretation: in characteristic 2 the equation is inseparable, while in characteristic 5 it is generically $z^2 = 0$.

This abstract integral algebra does not by itself give an integral incidence theorem. The constructions in the cited Mukai–Umemura and Hitchin sources are over $\mathbf{C}$, and the available binary-sextic formulae justify their classical invariant presentation only after inverting 30. Spreading out the rational incidence isomorphism gives some $\mathbf{Z}[1/N]$, with $2, 3, 5 \mid N$ after enlarging N , but neither the sources nor the present equations determine the other prime divisors or prove that there are none. Consequently every finite-field icosahedral-incidence interpretation in this paper remains restricted to an unspecified cofinite set of primes. The formula for the abstract quadratic algebra itself has no such extra exception.

The missing assertion is precise. One must identify an integral harmonic lattice and its three-skew-form Grassmannian zero locus with an integral Mukai–Umemura model, prove flatness and normality, and show that Stein formation and the chart sections commute with the required base changes. Until those geometric steps are supplied, extra divisors of N are evidence gaps rather than additional bad primes predicted by the quadratic equation.

3 The marked orientation source

This section keeps track of the sign lost by projectivizing the Clebsch chart. The normalization is intrinsic to the pullback. Its comparison with the golden six-axis data and the Petersen coefficient module is relative to the marked bridge datum $\mathbf{m}$ fixed in the introduction.

3.1 Normalization over the Clebsch chart

Let $\mathcal{N} \rightarrow \mathbf{P}(H)$ be the incidence normalization from Section 2, and put $B = \mathbf{P}(V_E)$. Choose the anti-invariant generator in Theorem 1.1; its pullback is the odd generator z below. On the chart, the Stein-algebra theorem and Hitchin’s restriction identity give the equality of finite $\mathcal{O}_B$ -algebras

$$z^2 = 5\iota_t^* J_0 = 80\sigma_3^2 = (z - 4\sqrt{5}\sigma_3)(z + 4\sqrt{5}\sigma_3).$$

For a normal domain S , with 2 invertible and $0 \neq a \in S$, the algebra $S[z]/(z^2 - a^2)$ maps to $S \times S$ by $z \mapsto (a, -a)$. Its image consists of pairs congruent modulo a ; hence its normalization is $S \times S$, its conductor is $(a) \times (a)$, and it is already split etale where a is a unit. Applying this elementary pinching lemma with $a = 4\sqrt{5}\sigma_3$ gives the normalization $B_+ \amalg B_-$ and locates the entire failure of normality on the Clebsch cubic. The constant plane U_t supplies one component; deck exchange supplies the other. Hitchin’s chart classification identifies $D(\sigma_3)$ as exactly the locus of two distinct regular configurations. The magnitude $4\sqrt{5}$ is forced by the product of the descent class 5 and the chart factor 16; choosing its sign is precisely choosing one normalized component.

The sign of z is the remaining freedom in the chosen anti-invariant generator. We normalize it by declaring the component supplied by U_t to have $z = 4\sqrt{5}\sigma_3$. This does not strengthen the integral statement: comparison with the geometric incidence model still holds only over an unspecified $\mathbf{Z}[1/N]$.

3.2 The golden involution

Choose the displayed normalized representatives $a_0, \dots, a_5$ of the ordered projective axes in $\mathfrak{m}$, and write $C = C_{\mathfrak{m}}$ and $Z_C = Z_{\mathfrak{m}}$. Their Gram matrix has the form

$$G = (t+2)I + tC, \quad C = \begin{pmatrix} 0 & 1 & 1 & 1 & -1 & -1 \\ 1 & 0 & -1 & -1 & -1 & -1 \\ 1 & -1 & 0 & 1 & 1 & -1 \\ 1 & -1 & 1 & 0 & -1 & 1 \\ -1 & -1 & 1 & -1 & 0 & -1 \\ -1 & -1 & -1 & 1 & -1 & 0 \end{pmatrix}.$$

The six vertex axes of an icosahedron as equiangular lines, and their signed Gram or Seidel-matrix description up to diagonal switching, are classical [12, §3]. What is tracked below is the sign of the specific marked triangle cubic under the golden exchanger. The frame operator $\sum_i a_i a_i^\top$ commutes with the irreducible three-dimensional A_5 -action, so it is scalar. Its trace is $6(t+2)$; division by the dimension gives $\sum_i a_i a_i^\top = 2(t+2)I_3$. Hence $G^2 = 2(t+2)G$. Substituting $G = (t+2)I + tC$ and using $(t+2)^2 = 5t^2$ gives $C^2 = 5I$. Changing signs of axis representatives switches C to DCD , but every triangle product and hence Z_C is unchanged. The off-diagonal entries of C^2 also give

$$\sum_{k \neq i, j} C_{ij} C_{jk} C_{ki} = 0.$$

Thus the directional derivative of Z_C along $\mathbf{1}$ vanishes, so $Z_C(x + u\mathbf{1}) = Z_C(x)$ and the cubic descends to $\mathbf{Q}^6/\mathbf{Q}\mathbf{1}$.

Let a'_i denote the conjugate axes, obtained by $t \mapsto 1-t$, and let R be the exchanger of Section 4. Direct substitution gives

$$Ra_i = td_i a'_{p(i)}, \quad p = (0, 1, 5, 4, 2, 3), \quad (d_i) = (1, -1, 1, 1, 1, 1).$$

Using $t^2(1-t) = -t$, one obtains

$$d_i d_j C_{p(i)p(j)} = -C_{ij}.$$

Thus, at the golden fibre, conjugation transported by R changes C to $-C$ and Z_C to $-Z_C$. The lone negative d_i is only a change of axis representatives and does not cause the reversal; switching alone leaves the triangle cubic fixed.

No global marking of the varying configurations on B_- is needed or asserted. The fixed section U_t selects B_+ , but does not by itself choose $\mathfrak{m}$. After $\mathfrak{m}$ is fixed, we attach its constant source $[C_{\mathfrak{m}}, Z_{\mathfrak{m}}]_{\text{or}}$ to B_+ and attach the deck-opposite source to B_- , meaning that the normalized odd generator changes sign. The calculation at $[xyz]$ proves that this relative convention agrees with the actual conjugate golden configuration there; it is not used to infer a pointwise conference labeling elsewhere. Since the normalized components remain disjoint above $\sigma_3 = 0$, the two relative source labels extend there even though the literal two-configuration interpretation fails.

3.3 Transport to degree six

Write Hitchin's chart as $\iota_t(y) = \sum_i y_i q_i^{(t)}$, with $q_1 = xyz$. Since $R(xyz) = -xyz$, equivariance and irreducibility of the four-module give

$$Rq_i^{(t)} = -q_i^{(1-t)}$$

after transporting the five plane-triple labels. Normalize the lift on B_+ by $q_1 = xyz$; the deck-opposite source carries the lift $-\iota_t$. The displayed exchanger calculation shows that this normalization agrees with golden conjugation at $[xyz]$. Thus the lift is part of the oriented source datum, rather than a purported global marking of the second family.

The primitive integral map

$$\beta(y)_{ij} = y_i + y_j$$

lands in the Petersen (-2) -eigenspace and satisfies

$$\sum_{i < j} (y_i + y_j)^2 = 3 \sum_i y_i^2, \quad y_i = \frac{1}{3} \sum_{j \neq i} \beta(y)_{ij}.$$

Multiplicity one makes this comparison unique after the five labels and the normalization are fixed. Reversing the chart lift reverses y , the zonal harmonic F_y , and its cubic integral. The two signs in the normalized incidence cover therefore select the two signs of the exact Gaunt cubic relative to $\mathbf{m}$. This proves Proposition 1.2.

The remaining marking actions and the boundary of this comparison are listed in Appendix A.

4 Arithmetic specialization

The arithmetic exchange is already visible on the cubic xyz . Its two icosahedral six-axis configurations are Galois conjugate over $\mathbf{Q}(\sqrt{5})$. The exchanger is the projective orthogonal transformation carrying one configuration to the other. Over a finite field where the configurations become rational, it has spinor class [2]; modulo 11 this class is nontrivial.

4.1 The golden fibre

Retain the field E , parameter t , and six-set I_t from Section 2. These six projective points are its axes through opposite vertices. Thus an icosahedral configuration here means the unordered six-set I_t . A point of the incidence cover records one chosen six-set; its two points over xyz correspond to the two conjugate configurations.

Proposition 4.1 (Golden fibre). *The points of I_t lie on $xyz = 0$, have a common nonzero norm, have normalized squared inner product $1/5$, and form a six-arc. The two configurations are conjugate geometric configurations defined over*

$$\mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5})$$

and are exchanged by its nontrivial automorphism. In Hitchin's incidence cover, whose point records the chosen six-set, they are the complete fibre over $[xyz]$.

Proof. Every displayed vector has one zero coordinate. Since $t^2 = t + 1$, its squared norm is $t^2 + 1 = t + 2$. The inner product of two distinct representatives is $\pm t$: the A_5 -action is transitive on pairs of distinct projective axes, so one squared inner-product check suffices. Moreover

$$(t + 2)^2 = 5(t + 1) = 5t^2.$$

This gives the normalized squared inner product $1/5$. The twenty three-point checks reduce to one signed Gram determinant. For any triple, let $\epsilon = \pm 1$ be the product of its three inner-product signs. Its Gram determinant is

$$(t + 2)^3 - 3(t + 2)t^2 + 2\epsilon t^3 = \begin{cases} 4t^4, & \epsilon = 1, \\ 4t^2, & \epsilon = -1. \end{cases}$$

It is nonzero, so every triple is independent. The final assertion uses Hitchin's classification of the two icosahedral sets on the orthogonal-plane cubic. $\square$

The six-arc conclusion remains valid after reduction in every characteristic other than 2. The metric interpretation additionally excludes characteristic 5, where $t^2 - t - 1$ is inseparable and the common norm $t + 2$ vanishes. This does not extend the complete-fibre assertion: the comparison with Hitchin's geometric incidence cover is available in characteristic zero and, after spreading out, only away from an unspecified finite set of primes.

4.2 The exchanger

Set

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The relation $1 - t = -1/t$ gives, by substitution,

$$R(I_t) = I_{1-t}, \quad R^2 = \text{diag}(1, -1, -1), \quad R^4 = 1.$$

Moreover $R(xyz) = -xyz$: it fixes the projective cubic and exchanges its two configurations.

4.3 Finite-field specialization

Let k be a finite field of odd characteristic other than 5, and suppose that 5 is a square in k . Then $t^2 - t - 1$ has two distinct roots in k , so the displayed equations give two rational configurations exchanged by R .

Proposition 4.2 (Spinor specialization). *The spinor class of R is $[2] \in k^\times/k^{\times 2}$. Under the standard identifications*

$$\text{SO}_3(k)/\Omega_3(k) \cong \text{PGL}_2(k)/\text{PSL}_2(k),$$

the exchanger lies outside $\text{PSL}_2(k)$ exactly when 2 is nonsquare. In particular, over $\mathbf{F}_{11}$ it represents the nontrivial element of

$$T_{11} = \text{PGL}_2(11)/\text{PSL}_2(11).$$

Proof. For $Q(x, y, z) = x^2 + y^2 + z^2$, reflection in a vector v has spinor class $Q(v)$. On the yz -plane,

$$R = s_{e_2} s_{e_2 - e_3},$$

so

$$\theta(R) = Q(e_2)Q(e_2 - e_3) = 2 \quad \text{in } k^\times/k^{\times 2}.$$

The standard identifications

$$\text{SO}_3(k) \cong \text{PGL}_2(k), \quad \Omega_3(k) \cong \text{PSL}_2(k)$$

identify the spinor quotient with the projective quotient. Finally, the roots modulo 11 are 4 and 8, and 2 is nonsquare modulo 11, proving the specialization. $\square$

The class [2] is forced by the orthogonal conjugacy class of this quarter-turn on the yz -plane; the two reflections merely evaluate its spinor norm.

Thus one equivariant quadratic datum carries two independent characters. The degree-two parameter algebra $k[t]/(t^2 - t - 1)$, with deck involution $t \mapsto 1 - t$, has discriminant class [5]; its two configurations are rational exactly when that torsor splits. The rational orthogonal element R realizes the deck involution on the configurations and has spinor character [2]. Dye's existence criterion for Clebsch hexagons has the same [5] boundary [3, Theorem 1].

Integral boundary. Over a finite field of characteristic different from 2 and 5, the fibre $\mathcal{Q}_f$ of the abstract quadratic cover $z^2 = 5J(f)$ over a rational point f gives

$$\#\mathcal{Q}_f(\mathbf{F}_q) = 1 + \chi_q(5J(f)).$$

Its geometric interpretation over finite fields is proved only away from an unspecified finite set of primes. This global comparison is separate from the good reduction at 11 of the explicit golden fibre and matrix R ; we do not replace the finite exceptional set by $p \neq 2, 5$.

5 The golden operator and its cubic shadows

The incidence cover supplies a golden orientation bit. Relative to the marked bridge datum, we attach that bit to the six-axis carrier and apply three elementary operations to its conference operator: middle exterior power, commutator with the diagonal algebra, and centered squaring. These operations produce the Joubert, Segre, Igusa, and diagonal Clebsch models in one calculation. The construction below makes the coherent operator family explicit. The theorem below proves that its apparently different cubic outputs are the same forms and then recognizes their classical quotient geometry.

Let X be the ordered six-axis set and $A_X = \mathbf{Q}^X/\mathbf{Q}\mathbf{1}$. The outer six-set $\mathcal{T}$ consists of the six synthemetic totals, that is, the six labelled one-factorizations of the fixed labelled graph K_X . These are the classical mystic pentagons. For each $T \in \mathcal{T}$, its complementary triangle colouring assigns signs $\epsilon_T(S) \in \{\pm 1\}$ to the three-subsets $S \subset X$, with the global sign fixed by the marked outer orientation. The signs obey the four-point two-graph identity and transform by

$$\epsilon_{gT}(gS) = \text{sgn}(g)\epsilon_T(S) \quad (g \in S_X).$$

This is the signed coloured-triangle construction of the outer six-family [13, Sections 1.1–1.2 and 1.6]. The order-six conference two-graph is likewise classical: its switching class has the pentagon-plus-isolated-vertex form and automorphism group A_5 [14]. The four-point identity reconstructs from each ϵ_T a switching class of symmetric zero-diagonal sign matrices; it is the unique order-six conference class. Choose representatives C_T whose triangle products are ϵ_T , and transport the Hodge orientations with the ordered axis basis. Then $C_T^2 = 5I$, and the resulting cubics defined below satisfy $gZ_T = \text{sgn}(g)Z_{gT}$. These choices constitute the coherent outer marking. The incidence sheet supplies only its orientation bit, not the outer labels or representatives.

The next theorem explains why this six-axis carrier has no cut-independent higher-order analogue. Let $C = C^\top$ be any symmetric conference matrix of order $2d$, that is, a zero-diagonal matrix with off-diagonal entries in $\{\pm 1\}$ and $C^2 = qI$, where $q = 2d - 1$. Its zero trace gives two eigenspaces of dimension d . For a balanced half Y , order the coordinates by (Y, Y^c) and write

$$C = \begin{pmatrix} A & R \\ R^\top & E \end{pmatrix}.$$

Let D_Y be the diagonal sign involution of this cut, put $P_{\pm} = (I \pm C/\sqrt{q})/2$, and choose an isometry $Q_+ : \mathbf{R}^d \rightarrow \text{im } P_+$. The balanced exchange operator is

$$H_Y = Q_+^{\top} D_Y P_- D_Y Q_+.$$

Theorem 5.1 (Balanced exchange rigidity). *With the notation above,*

$$\text{Spec}(H_Y) = \text{Spec}\left(\frac{1}{q} R R^{\top}\right) = \left\{1 - \frac{\alpha_1^2}{q}, \dots, 1 - \frac{\alpha_d^2}{q}\right\},$$

where $\alpha_1, \dots, \alpha_d$ are the eigenvalues of A . The spectrum is independent of the balanced half Y if and only if $d \leq 3$. Thus order six is the unique nontrivial realized symmetric conference order with this property, and its exchange spectrum is

$$\left\{\frac{1}{5}, \frac{4}{5}, \frac{4}{5}\right\}.$$

For a four-set K , let $w(K)$ be the sum of its three signed Hamilton-cycle products. Call K aligned if $w(K) = 3$, and let c_Y count the aligned four-sets in Y . Then

$$\text{tr}(H_Y) = \frac{d^2}{q}, \quad \text{tr}(H_Y^2) = \frac{F_d + 32c_Y}{q^2},$$

where

$$F_d = dq^2 - 2qd(d-1) + d(d-1) + 12\binom{d}{3} - 8\binom{d}{4}.$$

Hence the first exchange moment is universal, whereas the second is cut-independent exactly in the stated small orders.

Proof. Put $Q = C/\sqrt{q}$ and $L = [D_Y, Q]$. The commutator anticommutes with Q , and on its positive eigenspace one has $Lv = 2P_- D_Y v$. In cut coordinates,

$$L = \frac{2}{\sqrt{q}} \begin{pmatrix} 0 & R \\ -R^{\top} & 0 \end{pmatrix}.$$

Consequently $4H_Y$ is L^*L restricted to the positive eigenspace. Thus H_Y has the squared singular values of $R/\sqrt{q}$. The Y -principal block of $C^2 = qI$ is

$$R R^{\top} = qI - A^2,$$

which proves the spectral formula. Complementary principal-block spectral relations of this kind are classical [15].

Since A is a zero-diagonal sign matrix, $\text{tr}(A^2) = d(d-1)$, giving the first trace formula. Sorting the closed four-walks by support gives

$$\text{tr}(A^4) = d(d-1) + 12\binom{d}{3} + 8 \sum_{\substack{K \subset Y \\ |K|=4}} w(K).$$

Every edge occurs twice in the product of the three cycle signs, so $w(K) \in \{3, -1\}$. Substitution yields the displayed formula for $\text{tr}(H_Y^2)$. This is the exact finite version of the closed-walk organization used for conference-frame subensembles in [16].

Suppose now that $d \geq 4$ and even the second exchange moment is independent of Y . Then the sums of the aligned-four-set indicator over all balanced halves are equal.

The characteristic-zero inclusion matrix from four-sets to d -sets has full column rank [17, Theorem 1]; hence the indicator itself is constant on four-sets. Switch C so that all edges incident with one vertex ∞ are positive. On $\{\infty, i, j, k\}$, if the remaining edge signs are x, y, z , then

$$w = xy + xz + yz.$$

If the common value is 3, every triangle on the other $2d - 1$ vertices is monochromatic. All its edges have one sign, making the inner product of two corresponding conference rows equal to $2d - 2$, a contradiction. If the common value is -1 , that two-colouring has no monochromatic triangle; the elementary equality $R(3, 3) = 6$ gives $2d - 1 \leq 5$, again a contradiction.

For $d = 1$ the conclusion is immediate, and for $d = 2$ every possible two-by-two principal sign block squares to I . For $d = 3$, if τ is the product of the three edge signs, then

$$A^2 = 2I + \tau A,$$

and $\det(tI - A) = t^3 - 3t - 2\tau$. This polynomial factors as $(t - 2)(t + 1)^2$ or $(t + 2)(t - 1)^2$, so A^2 always has spectrum $\{4, 1, 1\}$. The spectral formula then gives $\{1/5, 4/5, 4/5\}$. $\square$

The obstruction is local, but the aligned sets retain uniform aggregate statistics. Greaves and Suda proved that the determinant- (-3) four-subsets of a symmetric conference matrix form a $3-(2d, 4, (d - 3)/2)$ design [18, Example 2.3]; the identity $\det C[K] = 3 - 2w(K)$ identifies their blocks with the aligned sets above. The regular-two-graph parameters give the same identification [19, Section 2.2 and Proposition 4.1]. Classical Johnson-scheme inclusion calculus [20, Section 8] therefore gives, for a uniformly chosen balanced half and $d \geq 4$,

$$\mathbf{E}c_Y = \rho \binom{d}{4}, \quad \text{Var}(c_Y) = \frac{\binom{2d}{4} \binom{2d-8}{d-4}}{\binom{2d}{d}} \rho(1 - \rho), \quad \rho = \frac{d - 3}{2(2d - 3)}.$$

These formulas also hold modulo complementation because $c_Y = c_{Y^c}$. At order ten they give $\mathbf{E}c_Y = 5/7$ and $\text{Var}(c_Y) = 10/49$, hence $\mathbf{E}[c_Y(c_Y - 1)] = 0$. Since c_Y is a nonnegative integer, it is therefore 0 or 1 on every cut. The 126 projective cuts consequently split into 36 with $c_Y = 0$ and 90 with $c_Y = 1$. The known design fixes this second-moment split without making the complete higher-order spectrum uniform.

A reverse design consequence. The design still retains the full discrete signing, despite fixing only aggregate low moments. Write a two-graph on a finite set V as a function

$$\tau : \binom{V}{3} \longrightarrow \mathbf{F}_2, \quad \sum_{S \in \binom{Q}{3}} \tau(S) = 0 \quad \left(Q \in \binom{V}{4} \right),$$

and let $\mathcal{A}(\tau)$ be the four-sets on which the four values of τ are all equal. Complementing τ preserves this family.

Theorem 5.2 (Aligned-design faithfulness). *If $|V| \geq 7$, then $\mathcal{A}(\tau)$ determines τ up to complement.*

For a symmetric conference matrix of order $n \geq 10$, its marked family of principal four-subsets with determinant -3 therefore determines its signing up to diagonal switching and global negation. After one aligned four-set Q is known, the

$$1 + 4(n - 4) + 6 \binom{n - 4}{2} = 3n^2 - 23n + 45$$

tests on four-sets meeting Q in at least two points already suffice. An aligned anchor is found deterministically with at most twenty tests, so the complete marked reconstruction uses $O(n^2)$ selected determinants. One calibrated triangle product selects between the two global orientations.

Proof. Fix $r \in V$ and form a graph G_r on $V \setminus \{r\}$, with ij an edge when $\tau(rij) = 1$. The two-graph equation gives

$$\tau(ijk) = \tau(rij) + \tau(rik) + \tau(rjk).$$

Thus $\{r, i, j, k\}$ is aligned exactly when ijk is a clique or an independent triple of G_r . The equality $R(3, 3) = 6$ supplies an aligned anchor on every seven-set.

It remains to prove faithfulness on seven points. Fix an aligned $Q = \{1, 2, 3, 4\}$, complement one candidate if necessary, and switch graph representatives so that both vanish on Q . Switch each outside vertex once more so that its edge to vertex 4 is zero. An outside point x then determines a uniquely normalized cut

$$p_x = (e_{1x}, e_{2x}, e_{3x}, e_{4x}) \in \mathbf{F}_2^4, \quad (p_x)_4 = 0.$$

The four tests meeting Q in three points have the following complete signature table; an entry ijk means that $\{i, j, k, x\}$ is aligned.

p_x	aligned triples	p_x	aligned triples
0000	123, 124, 134, 234	1000	234
0100	134	0010	124
1110	123	1100, 1010, 0110	$\emptyset$

Thus they recover p_x , except that the three balanced cuts $B_{12} = 1100$, $B_{13} = 1010$, and $B_{14} = 0110$ have the same empty signature.

We record the remaining finite step explicitly. For outside points x, y , put $p = p_x$, $s = p_y$, $e = e_{xy}$, and let $\mathcal{B}(p, s, e)$ be the set of pairs $ij \subset Q$ for which $\{i, j, x, y\}$ is aligned. Its four triangle values give

$$ij \in \mathcal{B}(p, s, e) \iff p_i + s_i = p_j + s_j \text{ and } e = p_j + s_i. \quad (5.1)$$

This criterion gives a complete classification. If p and s are already known, some pair has $p_i + s_i = p_j + s_j$, and exactly one value of e makes that pair aligned. If exactly one cut is balanced, relabelling Q reduces the known cut to 0000 or 1000; substituting the three balanced cuts in (5.1) gives

p	s	$\mathcal{B}(p, s, 0)$	$\mathcal{B}(p, s, 1)$
0000	B_{12}	34	12
0000	B_{13}	24	13
0000	B_{14}	14	23
1000	B_{12}	34	13, 14
1000	B_{13}	24	12, 14
1000	B_{14}	12, 13	23

so both the balanced cut and e are recovered. When both cuts are balanced, the remaining table is

$\{p, s\}$	$\mathcal{B}(p, s, 0)$	$\mathcal{B}(p, s, 1)$
$\{B_{12}, B_{12}\}$	12, 34	13, 14, 23, 24
$\{B_{13}, B_{13}\}$	13, 24	12, 14, 23, 34
$\{B_{14}, B_{14}\}$	14, 23	12, 13, 24, 34
$\{B_{12}, B_{13}\}$	23	14
$\{B_{12}, B_{14}\}$	13	24
$\{B_{13}, B_{14}\}$	12	34

The rows are distinct. The sole loss of information is the order of two distinct balanced cuts: interchanging p and s preserves the six pair tests and leaves e fixed.

There are three outside points. Suppose one cut changes between the two candidates. For each of the other two points the classification forces the pair to consist of distinct balanced cuts and swaps them. Those two other cuts are therefore equal to the same changed cut. Their mutual pair cannot be a distinct-cut swap, so it must agree pointwise, contradicting either of the first two swaps. No cut changes, and then (5.1) recovers every outside edge. The candidates agree after the initial possible complement.

For larger V , apply the seven-point result to every seven-set. Adjacent seven-sets in the connected Johnson graph share six vertices. On any triple in that intersection the two restrictions cannot choose opposite complement conventions, so their complement choices agree. This proves faithfulness. The same one-point and pair tests use exactly the displayed quadratic family. To find the anchor, fix a root and six other vertices and test the twenty four-sets formed by the root and a triple of the six; $R(3, 3) = 6$ makes at least one aligned.

For a conference signing, the identity $\det C[Q] = 3 - 2w(Q)$ identifies the determinant- (-3) family with $\mathcal{A}(\tau)$. Choosing a root and switching its incident signs to one recovers every matrix entry from the rooted triangle values. The sole remaining complement sends C to $-C$, and one triangle product fixes that orientation. $\square$

The closest reverse construction located is the homogeneous-triple reconstruction problem for ordinary graphs [23]. It supplies the one-anchor viewpoint but not the away-from-anchor tests that remove the balanced-cut ambiguity above. We have not located the aligned-design faithfulness theorem in the bounded audit; this sentence carries the same database qualifications as the singular-spectral statement below.

The neighboring framework is spectral monomorphy of principal submatrices [21, 22]. Here only the squared spectrum is assumed constant. We have not located the balanced singular-spectral classification above in that literature; the block identity, closed-walk expansion, inclusion rank, and aligned design are credited as indicated.

This exchange invariant deliberately forgets signs: it sees A^2 , or equivalently $RR^\top$. The cubic stage below retains the determinant-line orientation. Cut-independent squared singular spectrum, determinant orientation, and the later cubic shadows are therefore distinct claims.

For a three-subset $S \subset X$, write $x_S = \prod_{i \in S} x_i$, and set

$$K_T = * \bigwedge^3 C_T, \quad Z_T(x) = \frac{1}{4} \sum_{|S|=3} (K_T)_{SS} x_S, \quad D_x = \text{diag}(x).$$

The Hodge orientation is transported with the ordered axis basis. Thus an odd relabelling or switching transports both K_T and the sign of its diagonal; it is not legitimate to switch the matrix while holding the old Hodge convention fixed.

Theorem 5.3 (Golden operator and classical cubic shadows). *For a marked bridge datum and its coherent outer family:*

1. *The triangle products of C_T satisfy*

$$(K_T)_{SS} = 4(C_T)_{ij}(C_T)_{jk}(C_T)_{ki} \quad (S = \{i, j, k\}),$$

so Z_T is the switching-invariant orientation cubic on A_X . The four-point two-graph identity and pair balance recover the conference switching class; after fixing a vertex, the negative edges on the other five vertices form a pentagon.

2. The same cubic is a commutator Pfaffian:

$$\text{Pf}[D_x, C_T] = 4Z_T(x), \quad \det[D_x, C_T] = 16Z_T(x)^2.$$

Over $E = \mathbf{Q}(\sqrt{5})$, put $P_{T,\pm} = (I \pm C_T/\sqrt{5})/2$ and $B_T(x) = P_{T,-}D_xP_{T,+}$. After orienting the two determinant lines,

$$Z_T(x) = \pm 10\sqrt{5} \det B_T(x).$$

3. The six coordinates $(Z_T)_{T \in \mathcal{T}}$ satisfy

$$\sum_T Z_T = 0, \quad \sum_T Z_T^3 = 0.$$

They are the signed Joubert coordinates and hence define the classical map from six labelled points to the Segre cubic

$$\left\{ [z] \in \mathbf{P}^5 : \sum_T z_T = \sum_T z_T^3 = 0 \right\}.$$

Inside its hyperplane $z_T = 0$, the remaining linear relation cuts the ambient space to $\mathbf{P}^3$, and the remaining cubic equation is the diagonal Clebsch cubic surface.

4. The centered determinants

$$W_T = Z_T^2 - \frac{1}{6} \sum_U Z_U^2 = \frac{1}{16} \text{center}_T \det[D_x, C_T]$$

satisfy

$$\sum_T W_T = 0, \quad \left(\sum_T W_T^2 \right)^2 = 4 \sum_T W_T^4.$$

Thus W is the Segre–Igusa polar map.

Proof. The tight-frame argument in Section 3 gives $C_T^2 = 5I$. Its off-diagonal entries are the pair-balance identities

$$\sum_{k \neq i,j} (C_T)_{ij}(C_T)_{jk}(C_T)_{ki} = 0.$$

Triangle products are unchanged by diagonal switching and obey the four-point identity because every edge of a four-set occurs twice. In the gauge $(C_T)_{0i} = 1$, pair balance says that every vertex among the other five has two negative incident edges. The negative graph is therefore a five-cycle. This directly recovers the classical switching-class normal form without an enumeration.

Expand the Pfaffian of $[D_x, C_T]$ over perfect matchings. The coefficient of x_S pairs S bijectively with S^c , and its signed sum is the complementary 3-minor of C_T . The Hodge convention makes this minor $(K_T)_{SS}$. Two dihedral representatives and complementation give $(K_T)_{SS} = 4(C_T)_{ij}(C_T)_{jk}(C_T)_{ki}$, so the Pfaffian is $4Z_T$; squaring gives the determinant identity.

The projectors split the axis space into two three-spaces. Relative to that splitting,

$$[D_x, C_T] = \begin{pmatrix} 0 & -2\sqrt{5} B_T(x)^\top \\ 2\sqrt{5} B_T(x) & 0 \end{pmatrix}.$$

Comparison with $16Z_T^2$ gives $Z_T^2 = 500 \det(B_T)^2$; the chosen determinant-line orientation fixes the displayed sign.

Finally, by construction the coefficient of x_S in Z_T is the signed triangle colour $\epsilon_T(S)$. Thus the six operator cubics are exactly the signed coloured-triangle Joubert coordinates, including the normalization and transformation rule $gZ_T = \text{sgn}(g)Z_{gT}$. Their Segre relations are classical [13, Sections 2.1–2.2], so $\sum Z_T = \sum Z_T^3 = 0$. Setting one coordinate to zero gives $\sum_{U \neq T} Z_U = \sum_{U \neq T} Z_U^3 = 0$, the diagonal Clebsch cubic. The centered-square formula and the symmetric Igusa equation are the classical Segre–Igusa duality map [13, Section 2.2], following van der Geer. Here the determinant identity in the theorem identifies that polar vector with the centered commutator determinants. $\square$

Why the determinant. The map $B_T(x) : V_{T,+} \rightarrow V_{T,-}$ is defined without choosing bases. Its determinant is therefore a map between determinant lines and becomes the displayed scalar once those lines are oriented. A permanent has no analogous basis-free meaning. Indeed, changing orthonormal frames sends a matrix B to $R_-^T B R_+$; already for $B = I_3$, right multiplication by a rotation through angle θ in a coordinate plane changes its permanent from 1 to $\cos(2\theta)$. Thus the permanent supplies no further cubic shadow of the marked operator.

No physical or exceptional-group interpretation is used here. The retained chain is incidence descent, conference carrier, classical cubics, and the harmonic return of Section 6.

6 The degree-six harmonic realization

Here $\mathcal{H}_6$ denotes the real degree-six spherical harmonics on S^2 . This section is the harmonic return of the diagonal Clebsch invariant isolated by the operator theorem, not another operator shadow. The marked pair-sum map carries the relative orientation to the Petersen coefficient module, after which the construction passes from icosahedral face axes to $\mathcal{H}_6$.

Put $\phi = (1 + \sqrt{5})/2$. The following vectors have squared norm 3; their projective classes are the ten axes through opposite faces of an icosahedron:

$$\begin{aligned} v_{12} &= (-1, -1, -1), & v_{13} &= (-1, -1, 1), & v_{14} &= (-1, 1, -1), & v_{15} &= (-1, 1, 1), \\ v_{34} &= (0, -\phi^{-1}, -\phi), & v_{25} &= (0, -\phi^{-1}, \phi), & v_{45} &= (-\phi^{-1}, -\phi, 0), \\ v_{23} &= (-\phi^{-1}, \phi, 0), & v_{35} &= (-\phi, 0, -\phi^{-1}), & v_{24} &= (\phi, 0, -\phi^{-1}). \end{aligned} \tag{1}$$

We use the unit representatives $u_{ij} = v_{ij}/\sqrt{3}$. Two labels are adjacent when the pairs are disjoint. This is the Petersen graph $KG(5, 2)$, and the labeling is geometric because, for distinct axes,

$$(u_{ij} \cdot u_{kl})^2 = \begin{cases} 5/9, & \{i, j\} \cap \{k, l\} = \emptyset, \\ 1/9, & \text{otherwise.} \end{cases}$$

Replacing ϕ by its conjugate gives the conjugate embedded configuration. The abstract Petersen incidence and the formulas below are unchanged after the corresponding relabeling. We do not claim that this displayed embedding of its four-space is defined over $\mathbf{Q}$.

The zonal degree-six harmonics define

$$L : \mathbb{R}^{10} \longrightarrow \mathcal{H}_6, \quad L(a)(\omega) = \sum_{i < j} a_{ij} P_6(u_{ij} \cdot \omega),$$

where

$$P_6(s) = \frac{1}{16} (231s^6 - 315s^4 + 105s^2 - 5).$$

We equip $\mathcal{H}_6$ with the normalized spherical inner product

$$\langle f, g \rangle = \frac{1}{4\pi} \int_{S^2} f(\omega)g(\omega) d\omega.$$

Theorem 6.1 (Degree-six harmonic restriction). *Let u_{ij} , $1 \leq i < j \leq 5$, be the explicitly labelled unit axes through opposite icosahedral faces in (1), and set*

$$F_y(\omega) = \sum_{i < j} (y_i + y_j) P_6(u_{ij} \cdot \omega).$$

The coefficient vectors $(y_i + y_j)_{i < j}$ form the Petersen (-2) -eigenspace, and their zonal harmonics give an injective copy of V in the real degree-six spherical-harmonic space $\mathcal{H}_6$. The spherical cubic on this copy is

$$\frac{1}{4\pi} \int_{S^2} F_y(\omega)^3 d\omega = -\frac{784000}{1247103} \sigma_3(y).$$

6.1 Proof of Theorem 6.1

Let A be the Petersen adjacency matrix in the labeling (1). Direct evaluation gives the reproducing-kernel matrix

$$K_{ef} = P_6(u_e \cdot u_f) = \frac{1}{243} (196I + 47J - 112A).$$

Indeed $P_6(1) = 1$, while the two off-diagonal orbit values are $P_6(\sqrt{5}/3) = -65/243$ and $P_6(1/3) = 47/243$. Thus the three matrix coefficients contain no further data: $196 + 47 = 243$ on the diagonal and $47 - 112 = -65$ on Petersen edges. The addition theorem for spherical harmonics gives

$$G_{ef} = \langle P_6(u_e \cdot -), P_6(u_f \cdot -) \rangle = \frac{1}{13} K_{ef}.$$

Thus G , not K , is the Gram matrix for the stated inner product. Only two off-diagonal evaluations occur: A_5 is transitive on pairs of two-subsets with intersection zero and with intersection one. The three eigenvalues below therefore follow from the Petersen eigenvalues $3, -2, 1$, rather than from a ten-by-ten diagonalization. Its eigenvalues on $\mathbb{R}^{10} \simeq \mathbf{1} \oplus V_4 \oplus V_5$ are respectively

$$\frac{110}{1053}, \quad \frac{140}{1053}, \quad \frac{28}{1053}.$$

They are positive, so L is injective.

The Petersen eigenvalues are $3, -2, 1$ on $\mathbf{1}, V_4, V_5$. For $a_{ij} = y_i + y_j$ with $\sum_i y_i = 0$,

$$(Aa)_{ij} = \sum_{\{k,l\} \cap \{i,j\} = \emptyset} (y_k + y_l) = -2(y_i + y_j).$$

These vectors form the four-dimensional (-2) -eigenspace.

This identifies the abstract bridge uniquely. The action of A_5 on five letters is two-transitive, so an endomorphism of $\mathbf{Q}^5$ commuting with A_5 has one constant diagonal entry and one constant off-diagonal entry. Its restriction to V is therefore scalar. Consequently, once the five plane triples are identified, every A_5 -equivariant comparison between the arithmetic coordinate module and the Petersen coefficient module is a scalar multiple of

$$y \longmapsto (y_i + y_j)_{i < j}.$$

The inverse on its image is

$$y_i = \frac{1}{3} \sum_{j \neq i} a_{ij}.$$

Requiring the comparison to preserve σ_3 forces the scalar to have cube one, hence to equal one over $\mathbf{Q}(\sqrt{5})$. The abstract comparison is therefore unique with the stated labels and normalization. It still supplies no map between the two ambient harmonic spaces. The linear lift selected in Proposition 1.2 fixes its sign.

Nor can a rotation-equivariant polynomial covariant induce this linear bridge. Suppose such a map $\Phi : H_{\mathbb{R}} \rightarrow \mathcal{H}_6$ restricts to the displayed Clebsch four-space as the nonzero linear comparison above. Decomposing Φ into homogeneous parts and applying it to λy shows that its degree-one part would have the same restriction. But $\text{Hom}_{SO_3}(H_{\mathbb{R}}, \mathcal{H}_6) = 0$, because these are nonisomorphic irreducible rotation modules. Thus the marked A_5 -intertwiner is exactly the level at which a linear comparison can exist; no ambient rotation-equivariant polynomial covariant restricts to it.

The cubic integral is A_5 -invariant on V_4 . The invariant cubic line is generated by σ_3 , so one nonzero value determines the scalar. The vector $y = (4, -1, -1, -1, -1)$ spans the fixed line of a point stabilizer. Relative to the marked labels, this primitive vector fixes the scale and sign of the one-vector normalization. Its quadratic norm needs no new moment calculation: pair-sum tightness and the V_4 Gram eigenvalue give

$$\frac{\langle F_y, F_y \rangle}{\sum_i y_i^2} = 3 \left(\frac{140}{1053} \right) = \frac{140}{351}, \quad \frac{1}{4\pi} \int F_y^2 = \frac{2800}{351}.$$

Only the cubic normalization remains. Exact evaluation on this fixed line gives

$$\frac{1}{4\pi} \int F_y^3 = -\frac{15680000}{1247103}, \quad \sigma_3(y) = 20,$$

which proves the coefficient in Theorem 6.1. This one normalization value follows by expanding P_6 and applying

$$\frac{1}{4\pi} \int_{S^2} \omega_1^{2a} \omega_2^{2b} \omega_3^{2c} d\omega = \frac{(2a-1)!!(2b-1)!!(2c-1)!!}{(2a+2b+2c+1)!!};$$

monomials with an odd exponent integrate to zero.

The apparently large coefficient has two independent sources:

$$-\frac{784000}{1247103} = -\left(\frac{400}{46189} \right) \left(\frac{1960}{27} \right).$$

The first factor is $\left(\begin{smallmatrix} 6 & 6 & 6 \\ 0 & 0 & 0 \end{smallmatrix} \right)^2$. Thus $46189 = 11 \cdot 13 \cdot 17 \cdot 19$ is the universal degree-six Wigner denominator. The second factor is the one-vector restriction scalar after the unit-axis and primitive pair-sum normalizations have been fixed. Multiplicity one is what reduces the whole cubic calculation to this marked fixed line; $1960/27$ is therefore a normalization scalar, not an experimental coincidence. These denominator primes arise from the stated spherical and pair-sum normalizations and are unrelated to the incidence-cover square class. In the quadratic identity above, $3 = \|\beta(y)\|^2/\|y\|^2$, the second factor is the spherical Gram eigenvalue on V_4 , and the factor 13 in $1053 = 13 \cdot 81$ is $\dim \mathcal{H}_6$.

Remark 6.2. In the orthonormal Condon–Shortley convention, write $F = \sum_{m=-6}^6 q_{6m} Y_{6m}$ and

$$W_6(F) = \sum_{m_1+m_2+m_3=0} \begin{pmatrix} 6 & 6 & 6 \\ m_1 & m_2 & m_3 \end{pmatrix} q_{6m_1} q_{6m_2} q_{6m_3}.$$

The Gaunt formula and $\begin{pmatrix} 6 & 6 & 6 \\ 0 & 0 & 0 \end{pmatrix} = -20/\sqrt{46189}$ give

$$\int_{S^2} F^3 d\omega = -\frac{130}{\sqrt{3553}\pi} W_6(F).$$

This is the standard unnormalized degree-six bond-order invariant [6, 7, 8]. Combining this conversion with Theorem 6.1 gives the exact restriction in that standard language:

$$W_6(F_y) = \frac{313600\pi^{3/2}}{4563\sqrt{3553}} \sigma_3(y).$$

Here $3553 = 46189/13$ and $4563 = 27 \cdot 13^2$; both are forced by converting the universal Gaunt factor from the normalized spherical inner product to the standard Condon–Shortley convention.

7 Conclusion

Hitchin’s two conjugate Clebsch charts live over $\mathbf{Q}(\sqrt{5})$, where the branch sextic becomes $16\sigma_3^2$. Their common rational point $[xyz]$ lies in the finite étale locus of the rational incidence model. Their descended union has a nonsplit two-branch singularity there; its normalization fibre and conductor recover $\mathbf{Q}(\sqrt{5})/\mathbf{Q}$. The fixed-configuration sections map this normalization model canonically to the incidence normalization, whose complete reduced fibre determines the remaining rational square class. The explicit configurations also have a well-defined specialization at 11, without requiring a global integral incidence model there. Determining the exact finite set of primes that must be inverted for the geometric incidence comparison remains the integral identification, flatness, normality, and Stein-base-change problem isolated in Section 2; it is not a defect of the quadratic algebra $z^2 = 5J_0$.

After the outer marking and determinant-line orientations are fixed, the chosen incidence sign determines the oriented conference source on the six axes. Its tight frame forces $C^2 = 5I$; triangle holonomy, the diagonal of $*\bigwedge^3 C$, the commutator Pfaffian, and the cross-golden determinant are the same cubic. The six outer translates are the Joubert coordinates on the Segre cubic, their centered determinants are the Segre–Igusa polar coordinates, and a coordinate section is the diagonal Clebsch cubic surface. The cross-block calculation also characterizes the six-axis case: it is the unique nontrivial symmetric conference order whose balanced exchange spectrum is cut-independent. In higher orders the classical aligned-four-set design fixes the second-moment mean and variance but not the individual spectrum. Its marked blocks nevertheless recover every conference signing from order ten onward, up to switching and global negation, and quadratically many selected determinant tests suffice. One triangle calibration chooses the orientation used by the cubic shadows. Thus the conference stage remembers the incidence choice relative to exactly the marking data displayed in the theorem.

The degree-six face-axis construction returns to σ_3 : the normalized spherical cubic restricts to it on the Petersen four-space. After the golden axes, outer marking, plane triples, chart lift, and Petersen labels are fixed, the selected sheet is assigned the relative source $[C_m, Z_m]_{\text{or}}$; deck exchange gives its opposite, and the primitive Petersen map carries its sign to the degree-six Gaunt cubic. The sheet alone does not determine the marking or lift. This is a relative comparison of orientation sources, not an equality between Z_m and σ_3 or a map between their ambient harmonic spaces. The conclusion is therefore relative but exact: one marked golden source has four operator descriptions, its six outer coordinates recover the classical cubic models, and its Petersen image returns to the same invariant σ_3 .

A Marking ambiguities

The relative formulation uses the following complete action of the auxiliary choices.

operation	effect
axis switching	$C \mapsto DCD$, while every triangle product and Z is fixed; this is the equivalence already built into the marked source.
axis relabelling	simultaneously permutes the rows and columns of C and the variables of Z ; the proposition transports equivariantly and does not choose an order.
five-label relabelling	simultaneously permutes the q_i , the coordinates y_i , and the two-subset labels; σ_3 is fixed and β is equivariant. Relabelling only one side changes $\mathfrak{m}$.
chart scaling	rescales the linear identification and the transported cubic generator while leaving the projective chart fixed. It is not quotiented: $q_1 = xyz$ fixes the normalization. In particular, scaling by -1 reverses the generator.
golden Galois conjugation	after transport by R , sends C , Z , and the normalized chart lift to their negatives at the golden fibre.
incidence deck exchange	interchanges B_+ and B_- and negates the odd generator; relative to $\mathfrak{m}$, the attached source and harmonic generator are negated by definition.

Thus switching and coordinated relabelling do not change the comparison. Lift normalization and the cross-identification of the two five-labelled systems remain visible inputs. Proposition 1.2 makes no descent claim for them.

There is no nonzero rotation-equivariant linear comparison between harmonic degrees three and six: they are nonisomorphic irreducible rotation modules of dimensions seven and thirteen. The linear comparison used here is the unique marked A_5 -intertwiner on the chosen Clebsch four-space, and it stops at the relative marked source class $[C_{\mathfrak{m}}, Z_{\mathfrak{m}}]_{\text{or}}$.

B Reproducibility and data availability

The square-class argument and both specialization propositions are proved in the text from Hitchin’s incidence calculation and the displayed equations. A paper-local exact-arithmetic bundle independently checks the golden configurations, the exchanger, and its spinor decomposition. The human six-arc proof uses one Gram-determinant formula indexed by the triangle sign, so the stored twenty minors are regression data rather than twenty proof obligations.

A second paper-local bundle reconstructs the ten axes over $\mathbf{Q}(\sqrt{5})$, distinguishes the reproducing kernel from the normalized spherical Gram matrix, verifies the Petersen labeling, and evaluates the quadratic and cubic moments. Its independent replay uses a separate dense polynomial representation and checks the Wigner $3j$ conversion. Neither calculation uses floating-point arithmetic. The proof uses symmetry to reduce the kernel matrix to its two Petersen relations and uses only one point-stabilizer-fixed vector to normalize the invariant cubic line.

A third bundle checks, for the displayed marking, the conference square, the golden exchanger and its transport of C and Z_C , the scalar factorization of the pulled-back cover, and the primitive Petersen pair-sum identities. It reuses the canonical harmonic certificate only through its pinned digest and exact witness values. Normalization of the pulled-back incidence scheme and its extension across the branch divisor remain human arguments.

The operator theorem has a separate structural route. Its conference, triangle, two-graph, middle-exterior square and diagonal, support recovery, and golden-descent mechanisms are checked by the pinned golden-return Lean gate. The same gate proves symbolically that the fixed conference commutator Pfaffian is four times its triangle cubic and checks the three-vertex signed-matrix identity used in the order-six exchange-spectrum converse. The unrestricted inclusion-rank and Ramsey exclusion is a human proof. Lean also covers rooted recovery of triangle signs from a switched representative, but the seven-point aligned-design faithfulness argument, its Johnson-graph globalization, and its quadratic selected-query count are human proofs. Coherence of the outer six-family, the cross-golden determinant comparison, and the classical Joubert–Segre–Igusa identifications remain human proofs with primary literature inputs; they are not inferred from a finite matrix check.

The `verification` directory distributed with the paper contains the exact programs, canonical certificates, checksums, toolchain requirements, and aggregate replay command. It also records the exact claim-to-evidence correspondence. A paper-specific Lean gate now proves the abstract pinching and conductor mechanisms, involutive splitting, golden discriminant and reflection characters, tight-frame conference mechanism, marked relative sign transport, Petersen eigenspace and two-orbit kernel calculation, and fixed-line cubic normalization. Its declaration map records partial mechanism coverage only: no complete geometric manuscript row is declared formally proved at its full geometric strength. In particular, the global incidence and Stein identifications, scheme-theoretic chart correspondence, geometric golden-fibre identification, face-axis addition theorem, and raw spherical moment remain human inputs. The exact-arithmetic audits remain independent checks rather than hidden proof steps.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author’s direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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